Garrett Vernon

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U.S. Citizen | No Sponsorship Required | Eligible for Security Clearance | Open to Remote

SUMMARY

Software engineer building full-stack apps, REST APIs and AI/ML retrieval systems in Python, Java and TypeScript, deployed on AWS with CI/CD. Capstone team lead experienced in Agile sprints. B.S. earned fully online; M.S. Computer Science (AI) student in Georgia Tech's online OMSCS program.

TECHNICAL SKILLS

Languages: Python, SQL, Java, TypeScript, JavaScript, C#

Data & Databases: Relational databases and SQL: PostgreSQL (pgvector, full-text search), MySQL, SQLite, Redis, Firebase Firestore

Frameworks & APIs: FastAPI, Spring Boot, Spring Security, Next.js, Angular, Node.js, .NET MAUI; RESTful APIs, JWT/OAuth

AI / ML: PyTorch, LLM applications, retrieval-augmented generation (RAG), semantic search (pgvector), BM25, cross-encoder re-ranking, ONNX, LLM evaluation

Cloud & DevOps: AWS (Elastic Beanstalk, RDS, S3, CloudFront), Docker, Git, GitHub, GitHub Actions (CI/CD), Maven, npm, Vercel

Core Concepts: Agile/Scrum, Software Development Lifecycle (SDLC), RESTful web services, Object-Oriented Programming, Data Structures & Algorithms

EXPERIENCE

Heir Advisors Consulting, LLC January 2026 – May 2026
Software Engineering Intern

- Automated document-processing workflows across datasets of 100,000+ records, reducing repetitive manual work
- Built internal tooling for data extraction and ownership-research workflows
- Gathered requirements with leadership and delivered stakeholder-requested website enhancements

PROJECTS

Provenance — *Python, FastAPI, PostgreSQL/pgvector, PyTorch, ONNX*
AI question-answering system over SEC EDGAR filings that generates answers where every claim cites its exact source text.

- Engineered hybrid retrieval combining pgvector semantic search and BM25 full-text search, merged with Reciprocal Rank Fusion
- Fine-tuned an in-house cross-encoder re-ranker and exported it to ONNX for deployment
- Bootstrapped training data using an LLM, removing the need for manual labeling
- Designed an evaluation framework that measures answer faithfulness and citation accuracy

HomeLink Service Hub — *Java, Spring Boot, Angular, AWS (RDS, S3), GitHub Actions*
Deployed full-stack marketplace connecting homeowners with service providers through requests, job management and payments.

- Architected a three-role platform (homeowner, provider, administrator) with role-based access control (RBAC)
- Developed JWT-secured REST APIs using Spring Boot and integrated an Angular frontend
- Deployed cloud infrastructure on AWS (Elastic Beanstalk, RDS, S3, CloudFront) with CI/CD automation using GitHub Actions
- Independently rebuilt a university capstone into a deployed full-stack application

Cheevo — *Python, FastAPI, PostgreSQL, Redis, Docker, Next.js, TypeScript*
Full-stack Xbox achievement tracker integrated with Xbox Live to show game libraries, progress, rarity and Gamerscore.

- Developed background synchronization to keep game libraries and achievement progress up to date from an external API
- Designed a Redis caching layer to optimize external API requests and manage rate limits
- Integrated Xbox account authentication using OpenXBL OAuth with secure per-user token storage

Luex — *C#, .NET MAUI, Firebase Firestore: cross-platform financial tracking app for desktop and Android with real-time cloud sync*

EDUCATION

Georgia Institute of Technology *Expected Graduation: December 2028*
Master of Science, Computer Science (OMSCS, part-time) — AI Specialization

Franklin University, Columbus, OH May 2026
Bachelor of Science, Computer Science

- Team lead on capstone projects: planned sprint-based Agile work and kept the team on schedule
- Selected for Ohio's premier state-funded STEM merit award; re-qualified annually based on continuous academic excellence

CERTIFICATIONS

Python Programming Essentials, Rice University (Jan 2026) | Python Data Representations, Rice University (Jan 2026) | Mathematics for Machine Learning: Linear Algebra, Imperial College London (Apr 2026)